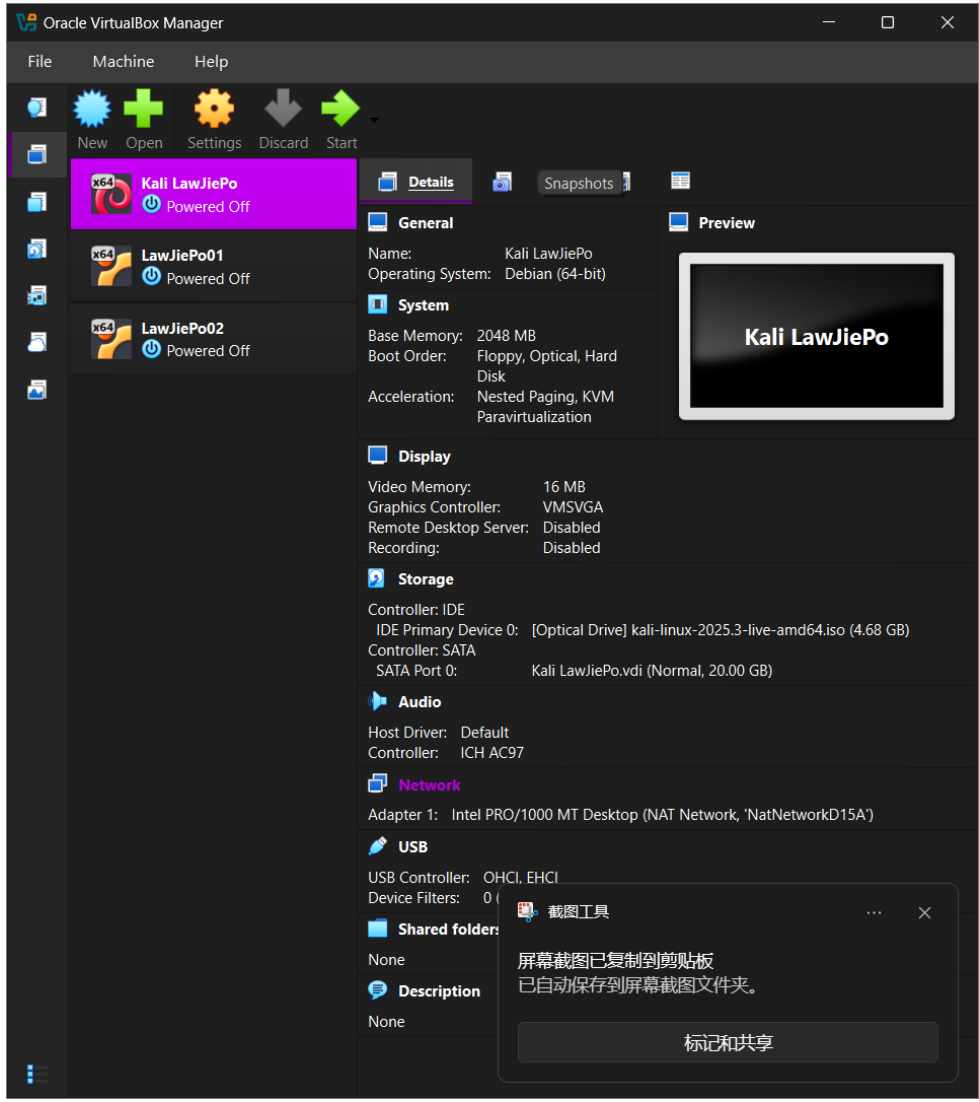
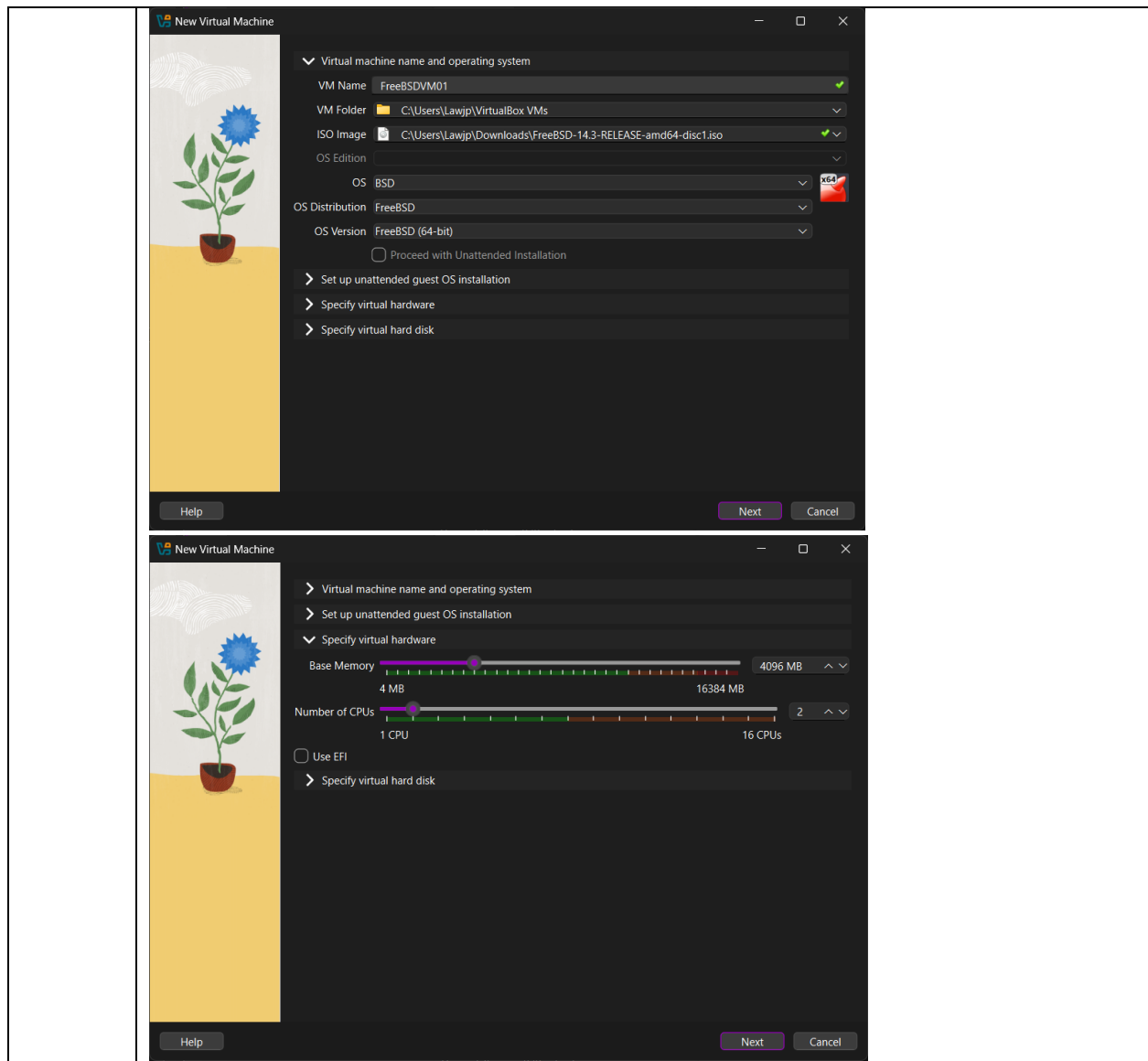


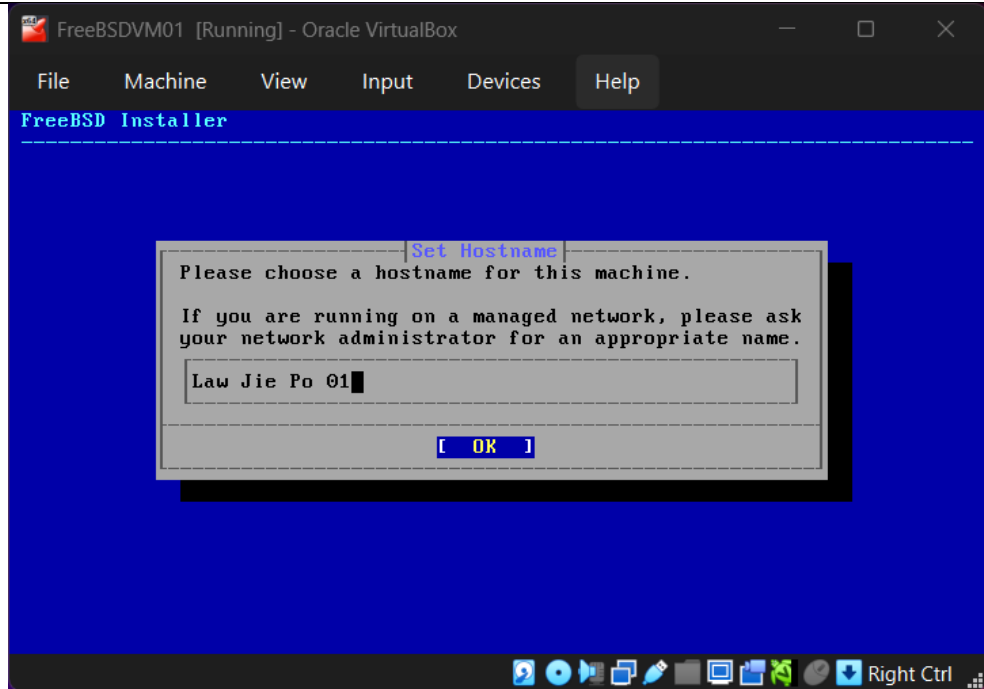
STUDENT ID	243DT245TU
NAME	Law Jie Po

No	Task	Host	Marks
1	All previous FreeBSD Virtual Machine must be deleted before lab test starts.	VirtualBox	2
2	Create 1 Virtual Machine based on FreeBSD. Name this machine FreeBSDVM01	VM01	8
3	Create VM02 Linux. Explain the name and version of your OS. Define suitable hostname.	VM02	10
4	Configure VM01 & VM02 to operate on the same network. Use Class C address.	VM01 VM02	10
5	Login as root in VM01	VM01	5
6	Show the current user in VM01	VM01	5
7	Show & explain the function 3 basic FreeBSD command in VM01	VM01	15
8	Create user (vmuser1) which can have root access	VM01	5
9	Create user (vmuser2) as normal user	VM01	5
10	Do a command to show FreeBSD version in VM01	VM01	5
11	Setup a web server in VM01(a website)	VM01	8
12	Edit index.html (website main page) with a short essay with a title "Server Virtualization" The short essay should have at least 40 words in multiple lines. Change the background page colour to blue.	VM01	5
13	Remote access VM01 from VM02 using vmuser2 account	VM01 VM02	5
14	Browse VM01 website from VM02	VM02	5
15	Properly power off VM01 & VM02	VM01 VM02	5
16	Snapshot your VirtualBox showing only VM01 & VM02	Laptop	2

Answer the following questions.

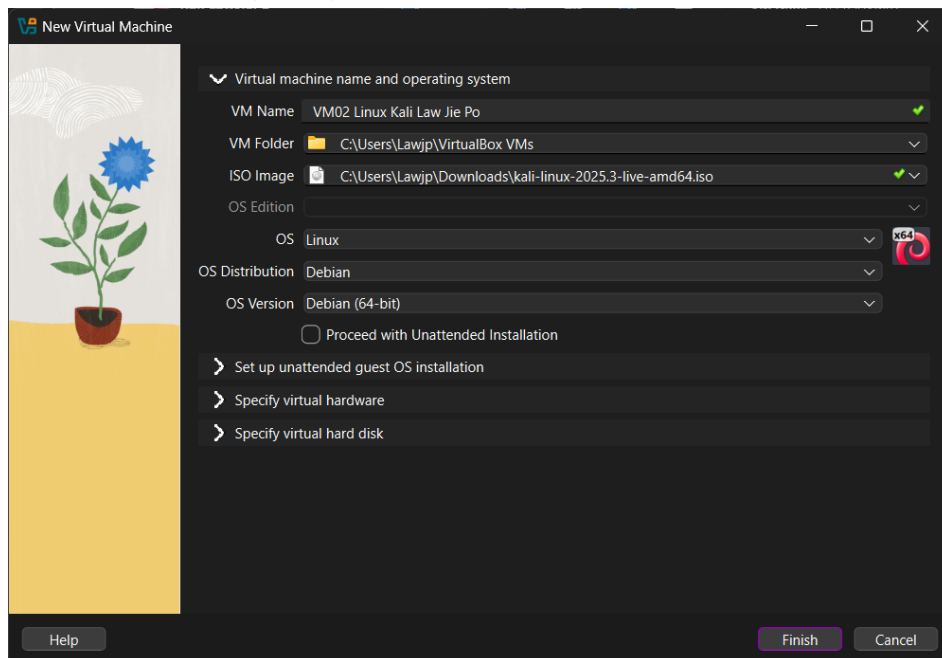
Question No.	Snapshot/ Answer
1) All previous FreeBSD Virtual Machine must be deleted before lab test starts. Snapshot your VirtualBox showing no FreeBSD virtual machine – 2m	
i.	
2) Hostname FreeBSD (VM01) – 8m	
i	Hostname: Law Jie Po 01

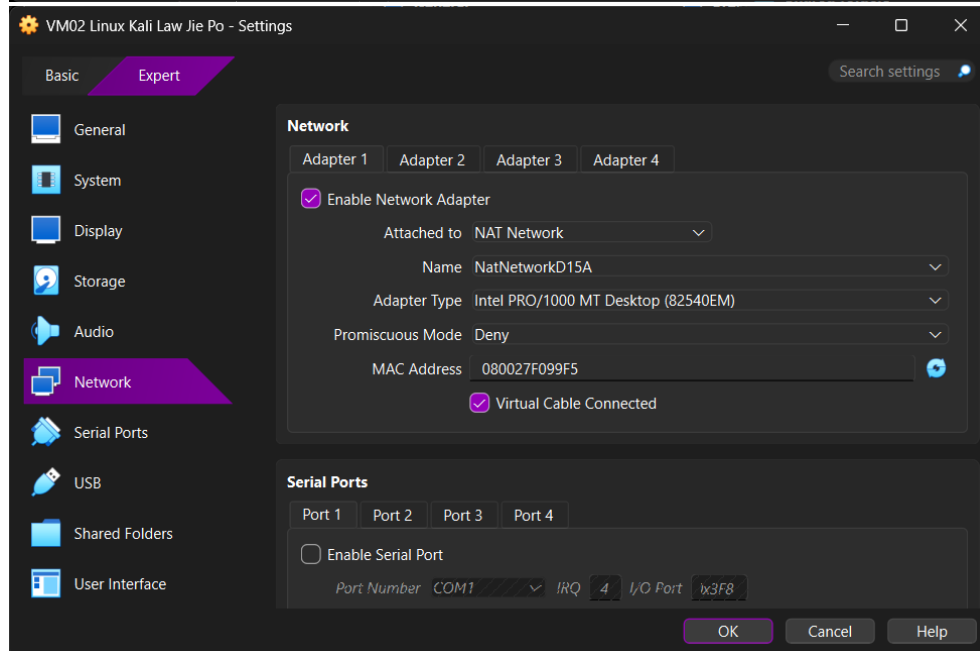
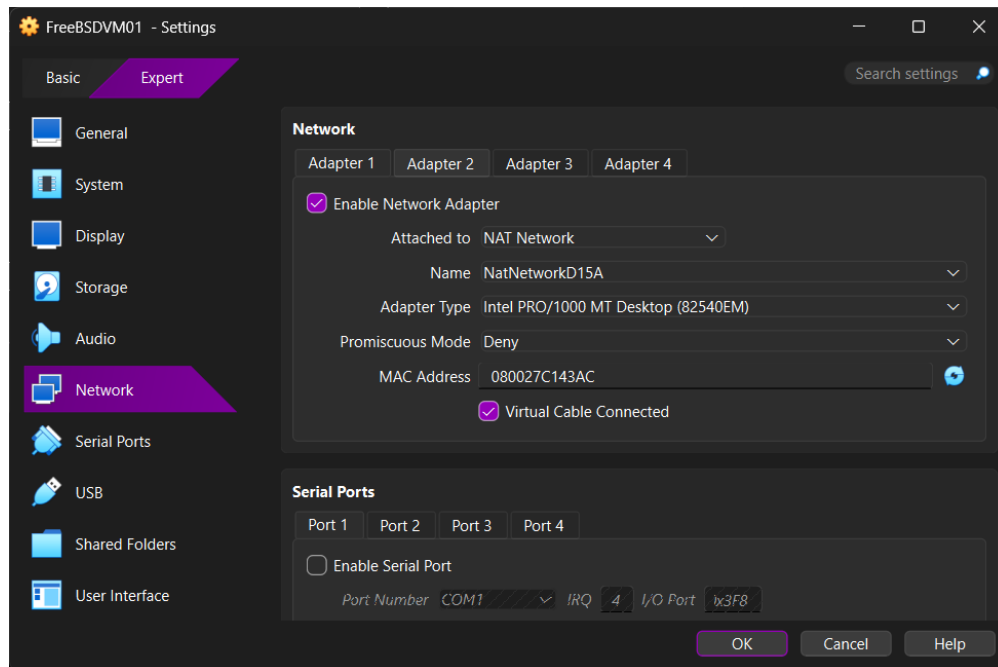


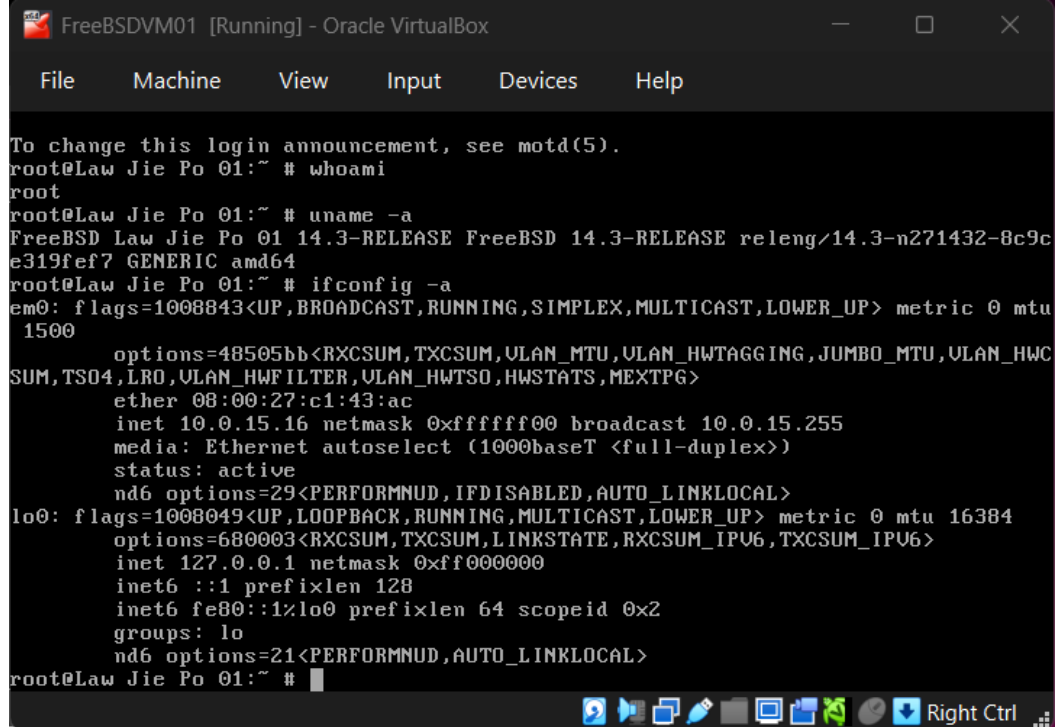


3) Hostname Linux (VM02) – 10m

- i. Hostname : VM02 Linux Kali Law Jie Po
OS Version: Debian (64-bit)

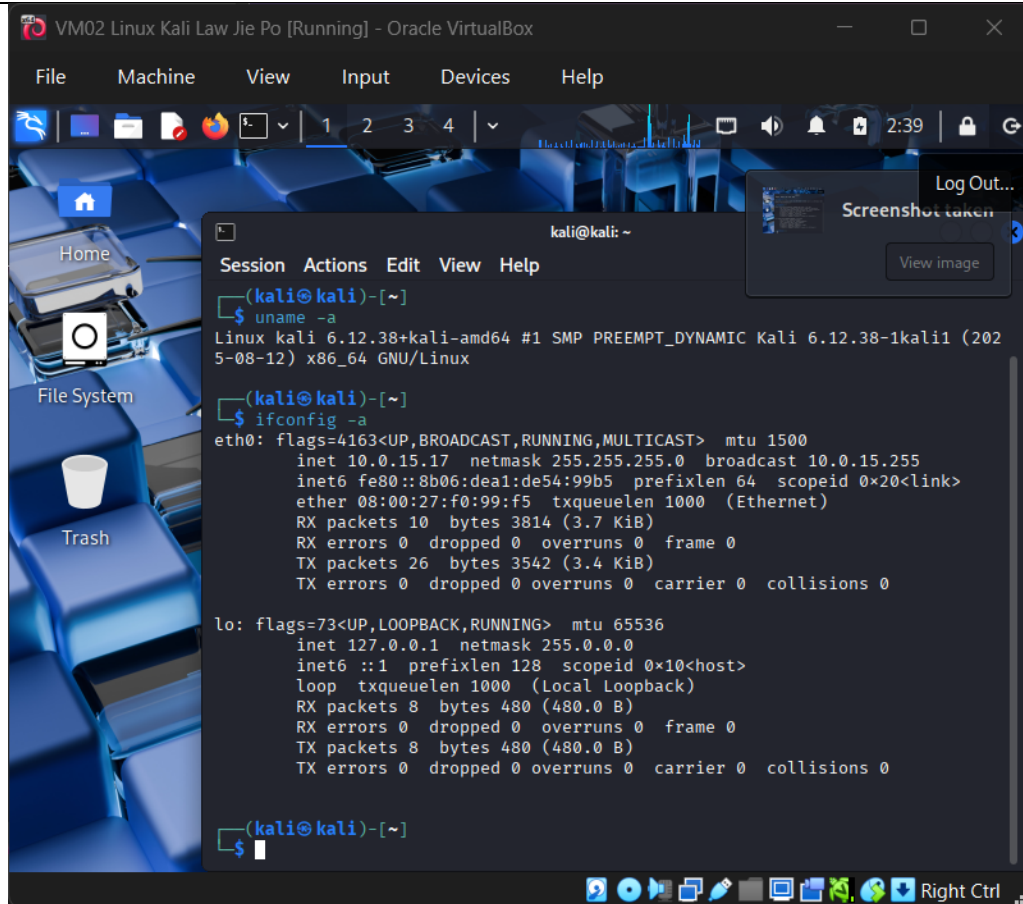


4) Configure VM01 & VM02 to operate in same network -10m**i.****IP Address VM01: 10.0.15.16****IP Address VM02: 10.0.15.17****Ip address of FreeBSD: 10.0.15.16**



```
To change this login announcement, see motd(5).
root@Law Jie Po 01:~ # whoami
root
root@Law Jie Po 01:~ # uname -a
FreeBSD Law Jie Po 01 14.3-RELEASE FreeBSD 14.3-RELEASE releng/14.3-n271432-8c9c
e319fef7 GENERIC amd64
root@Law Jie Po 01:~ # ifconfig -a
em0: flags=1008843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mtu
1500
options=48505bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMBO_MTU, VLAN_HWC
SUM, TS04, LRO, VLAN_HWFILTER, VLAN_HWTSO, HWSTATS, MEXTPG>
ether 08:00:27:c1:43:ac
inet 10.0.15.16 netmask 0xfffff00 broadcast 10.0.15.255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD, IFDISABLED, AUTO_LINKLOCAL>
lo0: flags=1008049<UP, LOOPBACK, RUNNING, MULTICAST, LOWER_UP> metric 0 mtu 16384
options=680003<RXCSUM, TXCSUM, LINKSTATE, RXCSUM_IPV6, TXCSUM_IPV6>
inet 127.0.0.1 netmask 0xff000000
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
groups: lo
nd6 options=21<PERFORMNUD, AUTO_LINKLOCAL>
root@Law Jie Po 01:~ #
```

Ip address of FreeBSD: 10.0.15.17

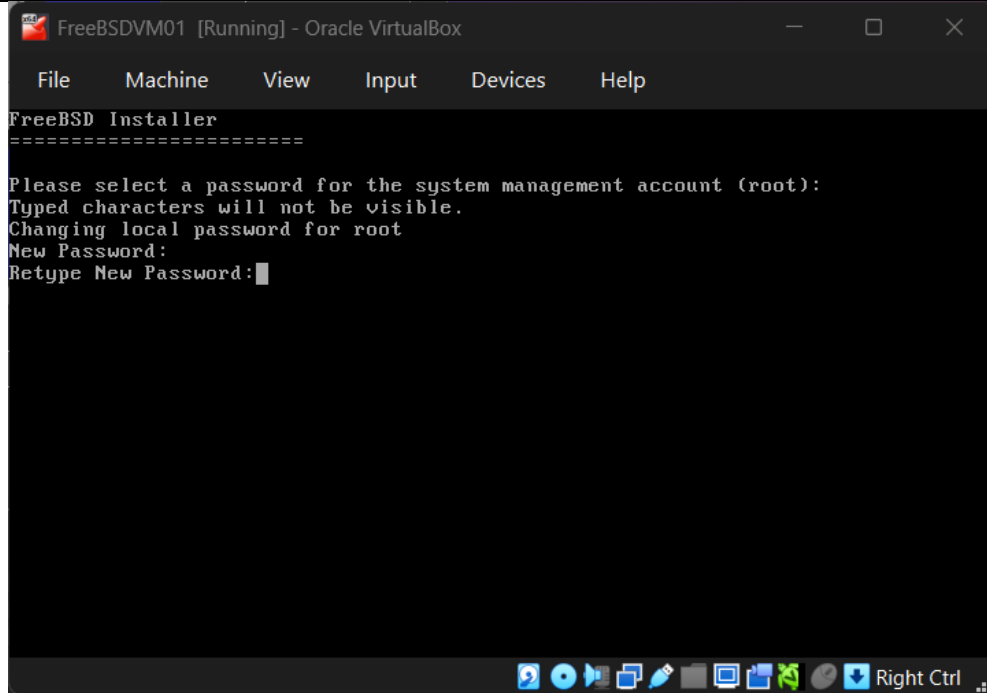


```
VM02 Linux Kali Law Jie Po [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Home
File System
Trash
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ uname -a
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64 GNU/Linux
(kali@kali)-[~]
$ ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.15.17 netmask 255.255.255.0 broadcast 10.0.15.255
    inet6 fe80::8b06:dea1:de54:99b5 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:f0:99:f5 txqueuelen 1000 (Ethernet)
    RX packets 10 bytes 3814 (3.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 26 bytes 3542 (3.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali@kali)-[~]
$
```

5) Login as root in VM01 – 5m

i. Password for root account: abc123



```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

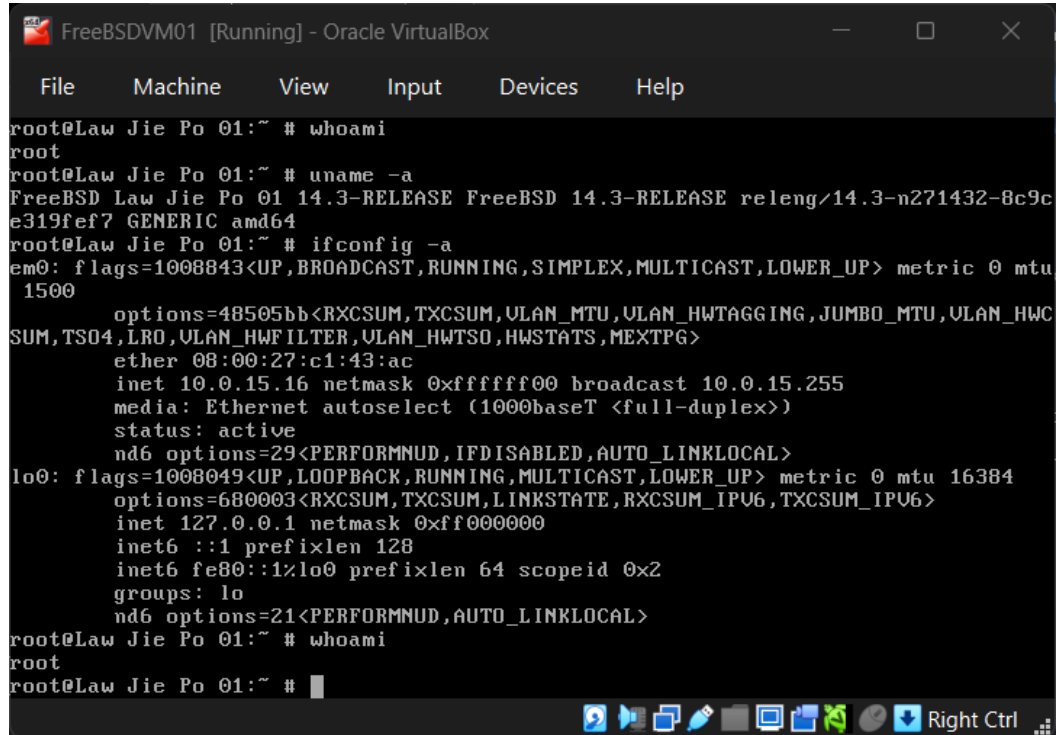
FreeBSD Installer
=====

Please select a password for the system management account (root):
Typed characters will not be visible.
Changing local password for root
New Password:
Retype New Password:
```

6) Show the current user in VM01 – 5m

i. Snapshot of file which shows VM01 users.

Current users here is root



```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@Law Jie Po 01:~ # whoami
root
root@Law Jie Po 01:~ # uname -a
FreeBSD Law Jie Po 01 14.3-RELEASE FreeBSD 14.3-RELEASE releng/14.3-n271432-8c9c
e319fef7 GENERIC amd64
root@Law Jie Po 01:~ # ifconfig -a
em0: flags=1008843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST,LOWER_UP> metric 0 mtu
1500
options=48505bb<RXCSUM, TXCSUM, VLAN_MTU, VLAN_HWTAGGING, JUMBO_MTU, VLAN_HWC
SUM, TS04, LRO, VLAN_HWFILTER, VLAN_HWTSO, HWSTATS, MEXTPG>
ether 08:00:27:c1:43:ac
inet 10.0.15.16 netmask 0xfffff00 broadcast 10.0.15.255
media: Ethernet autoselect (1000baseT <full-duplex>)
status: active
nd6 options=29<PERFORMNUD, IFDISABLED, AUTO_LINKLOCAL>
lo0: flags=1008049<UP, LOOPBACK, RUNNING, MULTICAST, LOWER_UP> metric 0 mtu 16384
options=680003<RXCSUM, TXCSUM, LINKSTATE, RXCSUM_IPV6, TXCSUM_IPV6>
inet 127.0.0.1 netmask 0xff000000
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
groups: lo
nd6 options=21<PERFORMNUD, AUTO_LINKLOCAL>
root@Law Jie Po 01:~ # whoami
root
root@Law Jie Po 01:~ #
```


7) Show & explain the function 3 basic FreeBSD command in VM01 -15m**i. Snapshot & explanation the 3 basic commands that you choose**

Whoami: to check I am user or the root in FreeBSD server

Uname -a: to check the version of FreeBSD

Ifconfig -a: to get the ipaddress of FreeBSD

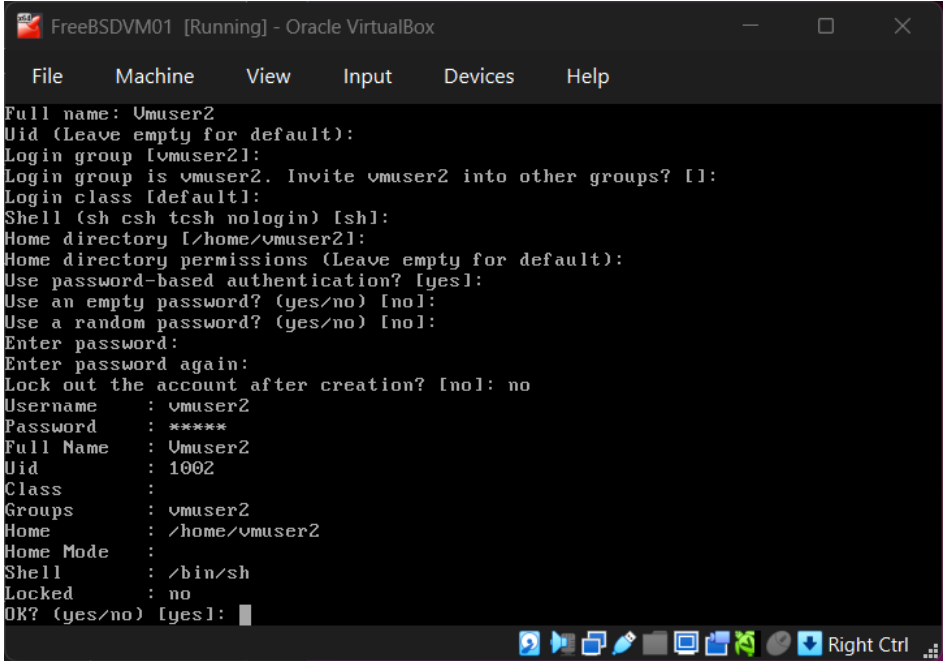
8) Create user (vmuser1) which can have root access – 5m	
i.	Snapshot the commands that you use <u>Code: adduser</u> <u>Login group need key in the wheel, then the vmuser1 can have root access</u>

```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

lo0: flags=1008049<UP,LOOPBACK,RUNNING,MULTICAST,LOWER_UP> metric 0 mtu 16384
      options=680003<RXCSUM,TXCSUM,LINKSTATE,RXCSUM_IPU6,TXCSUM_IPU6>
      inet 127.0.0.1 netmask 0xff000000
      inet6 ::1 prefixlen 128
      inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
      groups: lo
      nd6 options=21<PERFORMNUD,AUTO_LINKLOCAL>
root@Law Jie Po 01:~ # whoami
root
root@Law Jie Po 01:~ # adduser
Username: vmuser1
Full name: Vmuser1
Uid (Leave empty for default):
Login group [vmuser1]: wheel
Login group is wheel. Invite vmuser1 into other groups? [1]:
Login class [default]:
Shell (sh csh tcsh nologin) [sh]:
Home directory [/home/vmuser1]:
Home directory permissions (Leave empty for default):
Use password-based authentication? [yes]:
Use an empty password? (yes/no) [no]:
Use a random password? (yes/no) [no]:
Enter password:
Enter password again:
Lock out the account after creation? [no]:
```

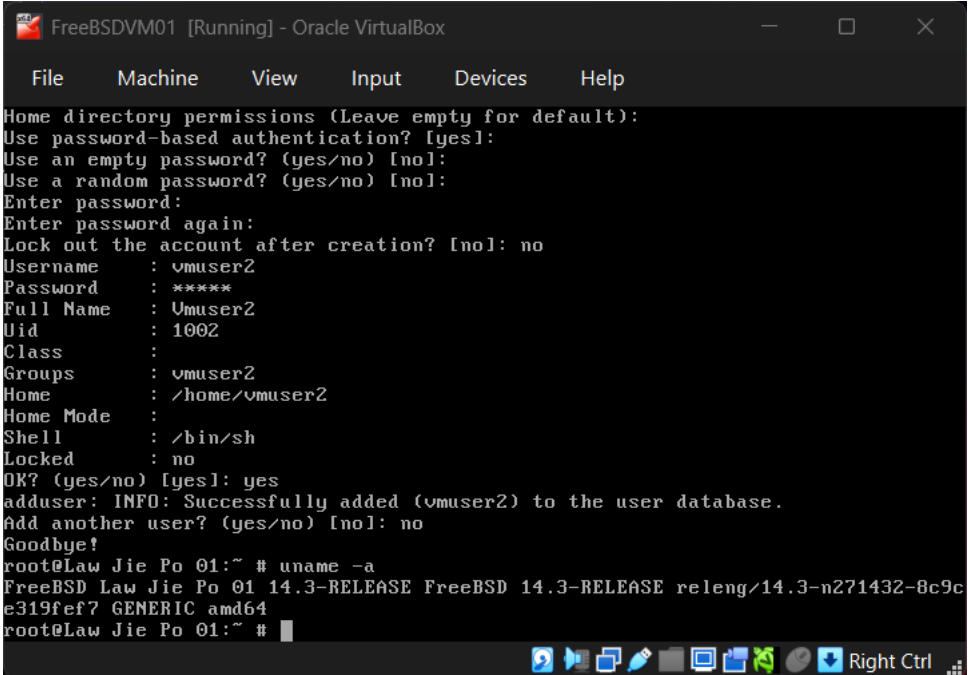
```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Login class [default]:
Shell (sh csh tcsh nologin) [sh]:
Home directory [/home/vmuser1]:
Home directory permissions (Leave empty for default):
Use password-based authentication? [yes]:
Use an empty password? (yes/no) [no]:
Use a random password? (yes/no) [no]:
Enter password:
Enter password again:
Lock out the account after creation? [no]: no
Username      : vmuser1
Password      : *****
Full Name     : Vmuser1
Uid           : 1001
Class         :
Groups        : wheel
Home          : /home/vmuser1
Home Mode     :
Shell         : /bin/sh
Locked        : no
OK? (yes/no)  [yes]: yes
adduser: INFO: Successfully added (vmuser1) to the user database.
Add another user? (yes/no) [no]: no
Goodbye!
root@Law Jie Po 01:~ #
```

9) Create user (vmuser2) as normal user – 5m	
i.	Snapshot the commands that you use Code: adduser Name: vmueser2 No need put something in login group, let this user become a normal user 

10) Do a command to show FreeBSD version in VM01 -5m

i.

Snapshot the commands that you use**uname -a use this to show FreeBSD version in VM01**

```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Home directory permissions (Leave empty for default):
Use password-based authentication? [yes]:
Use an empty password? (yes/no) [no]:
Use a random password? (yes/no) [no]:
Enter password:
Enter password again:
Lock out the account after creation? [no]: no
Username : vmuser2
Password : *****
Full Name : Vmuser2
Uid : 1002
Class :
Groups : vmuser2
Home : /home/vmuser2
Home Mode :
Shell : /bin/sh
Locked : no
OK? (yes/no) [yes]: yes
adduser: INFO: Successfully added (vmuser2) to the user database.
Add another user? (yes/no) [no]: no
Goodbye!
root@Law Jie Po 01:~ # uname -a
FreeBSD Law Jie Po 01 14.3-RELEASE FreeBSD 14.3-RELEASE releng/14.3-n271432-8c9c
e319fef7 GENERIC amd64
root@Law Jie Po 01:~ #
```

11) Setup a web server in VM01(a website) - 8m	

i.

List down the command that you do. Explain each step

Webserver: apache24

First need to correct the (date), code: date (to check/ change the date and confirm the date)

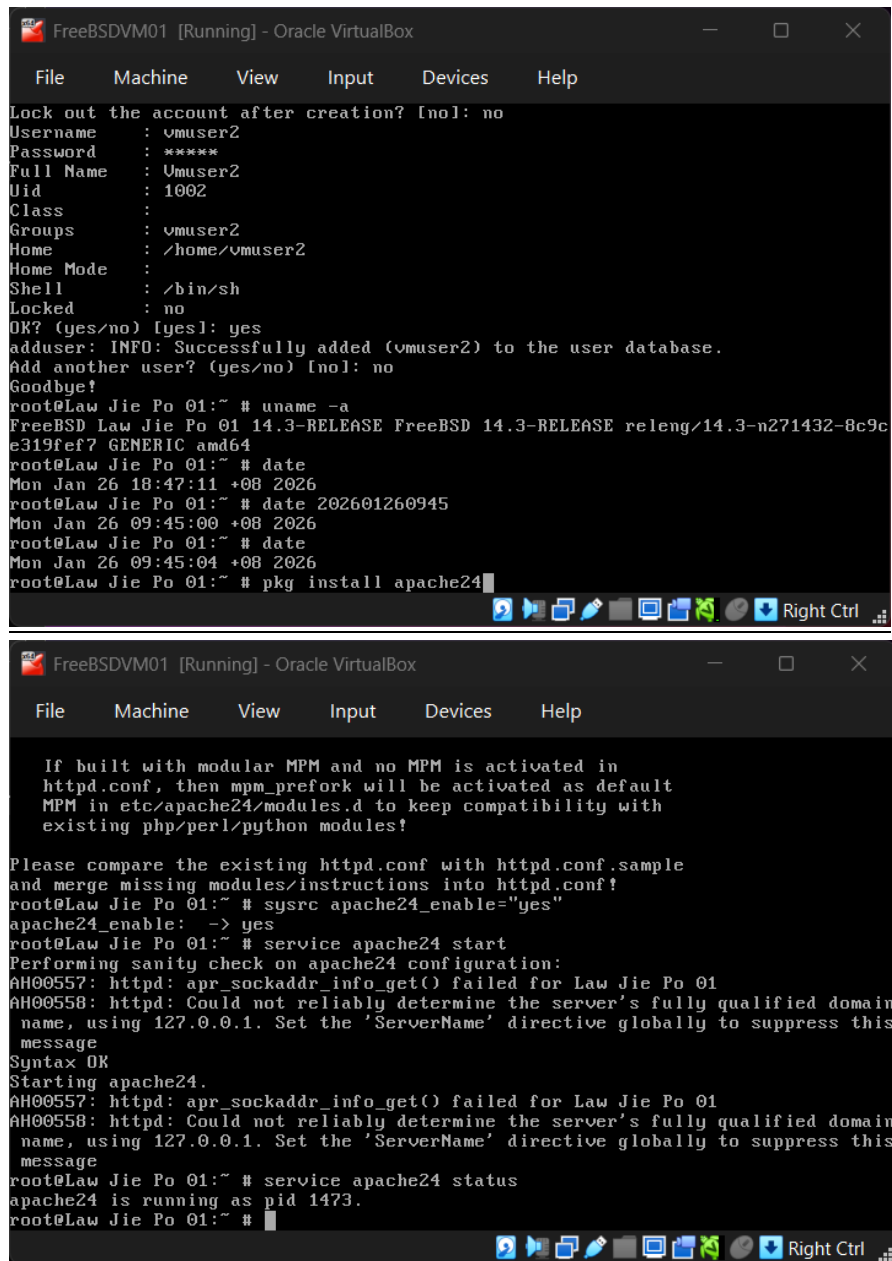
Second, need to use (pkg install apache24)

When installation needs to put two-time y(yes) to confirm install apache24

Now use (sysrc apache24_enable = "yes") to active the webserver in FreeBSD

Next need to start the apache24: (service apache24 start)

Last check the status of apache24 to confirm the apache24 have run or not: (service apache24 status)



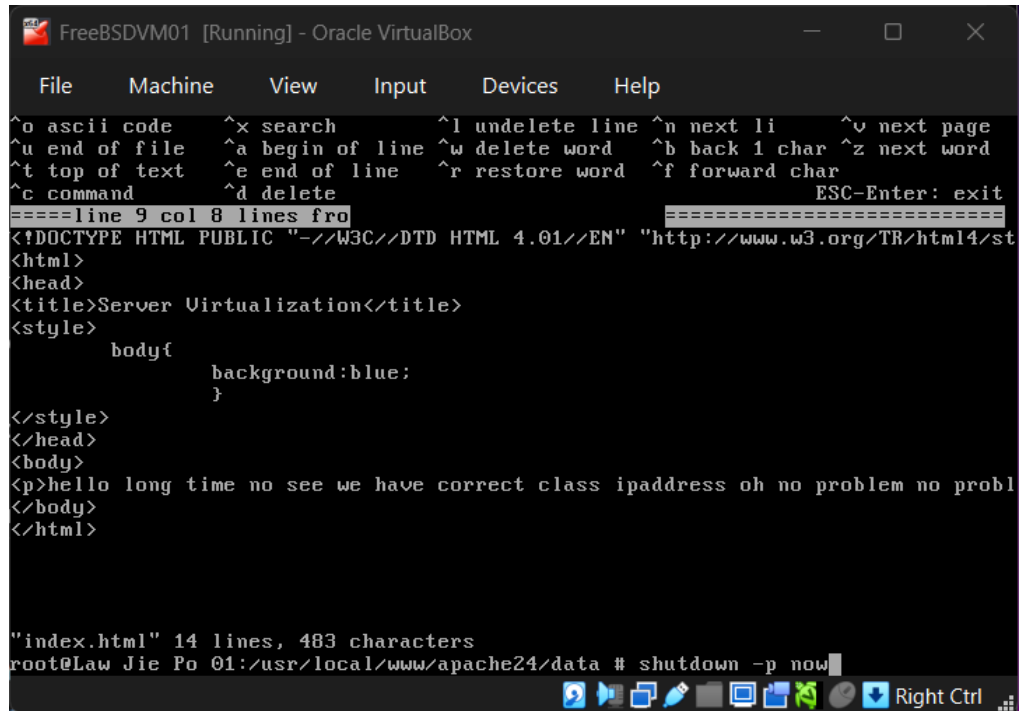
```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Lock out the account after creation? [no]: no
Username : vmuser2
Password : *****
Full Name : vmuser2
Uid : 1002
Class :
Groups : vmuser2
Home : /home/vmuser2
Home Mode :
Shell : /bin/sh
Locked : no
OK? (yes/no) [yes]: yes
adduser: INFO: Successfully added (vmuser2) to the user database.
Add another user? (yes/no) [no]: no
Goodbye!
root@Law Jie Po 01:~ # uname -a
FreeBSD Law Jie Po 01 14.3-RELEASE FreeBSD 14.3-RELEASE releng/14.3-n271432-8c9c
e319fef? GENERIC amd64
root@Law Jie Po 01:~ # date
Mon Jan 26 18:47:11 +08 2026
root@Law Jie Po 01:~ # date 202601260945
Mon Jan 26 09:45:00 +08 2026
root@Law Jie Po 01:~ # date
Mon Jan 26 09:45:04 +08 2026
root@Law Jie Po 01:~ # pkg install apache24

If built with modular MPM and no MPM is activated in
httpd.conf, then mpm_prefork will be activated as default
MPM in etc/apache24/modules.d to keep compatibility with
existing php/perl/python modules!

Please compare the existing httpd.conf with httpd.conf.sample
and merge missing modules/instructions into httpd.conf!
root@Law Jie Po 01:~ # sysrc apache24_enable="yes"
apache24_enable: -> yes
root@Law Jie Po 01:~ # service apache24 start
Performing sanity check on apache24 configuration:
AH00557: httpd: apr_sockaddr_info_get() failed for Law Jie Po 01
AH00558: httpd: Could not reliably determine the server's fully qualified domain
name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this
message
Syntax OK
Starting apache24.
AH00557: httpd: apr_sockaddr_info_get() failed for Law Jie Po 01
AH00558: httpd: Could not reliably determine the server's fully qualified domain
name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this
message
root@Law Jie Po 01:~ # service apache24 status
apache24 is running as pid 1473.
root@Law Jie Po 01:~ #
```

- 3) Edit index.html (website main page) with a short essay with a title "Server Virtualization". The short essay should have at least 40 words in multiple lines. Change the background page colour to blue. - 5m

- i. Snapshot the index.html file once you done the editing
Use code: ee (html file name) to edit the html file



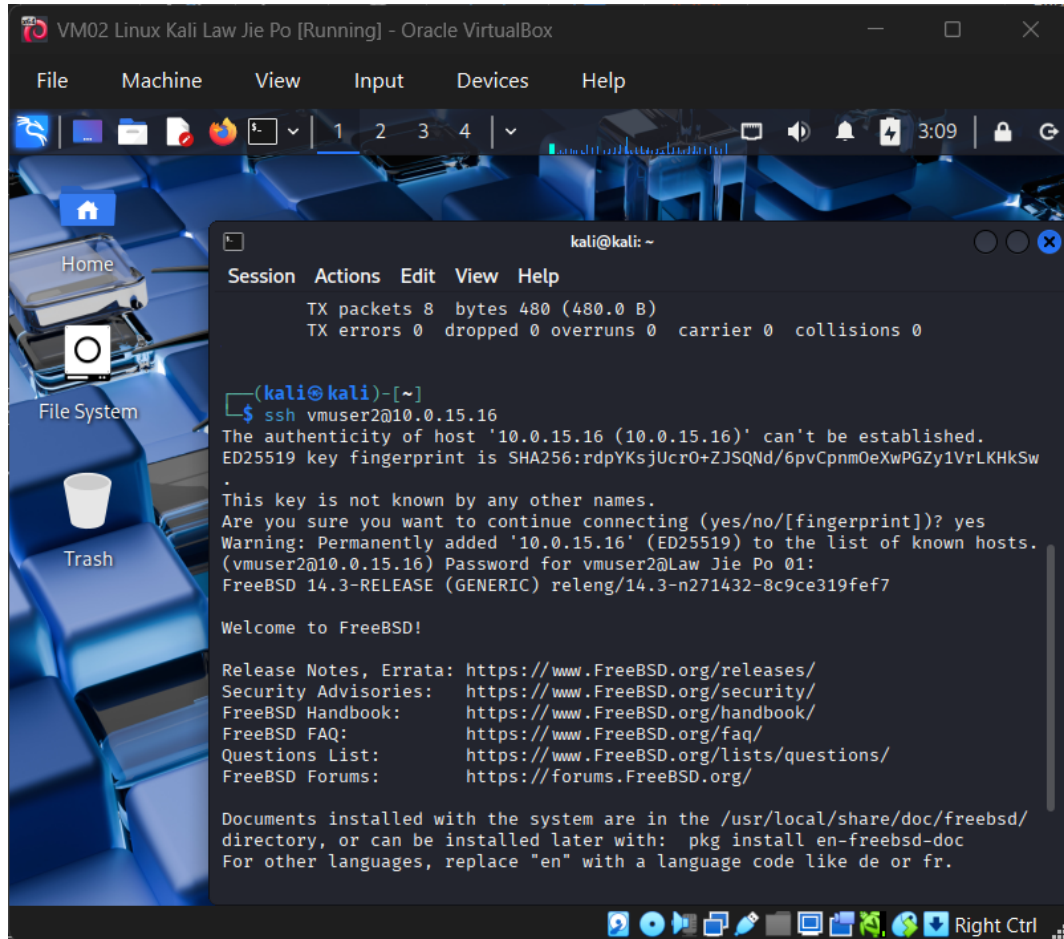
```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
^o ascii code ^x search ^l undelete line ^n next li ^v next page
^u end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete ESC-Enter: exit
=====line 9 col 8 lines fro=====
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/st
<html>
<head>
<title>Server Virtualization</title>
<style>
  body{
    background:blue;
  }
</style>
</head>
<body>
<p>hello long time no see we have correct class ipaddress oh no problem no probl
</body>
</html>

"index.html" 14 lines, 483 characters
root@Law Jie Po 01:/usr/local/www/apache24/data # shutdown -p now
```


4) Remote access VM01 from VM02 using vmuser2 account– 5m**i. Snapshot the commands that you use**

Use the command `ssh vmuser2@ipaddress(10.0.15.16)`

For Ip address need to check in vm01 `ifconfig -a` to get the Ip address and then can connect to vmuser2 in vm01



```
VM02 Linux Kali Law Jie Po [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali@kali)-[~]
$ ssh vmuser2@10.0.15.16
The authenticity of host '10.0.15.16 (10.0.15.16)' can't be established.
ED25519 key fingerprint is SHA256:rdpYKsjUcrO+ZJSQNd/6pvCpnmOeXwPGZy1VrLKHkSw
.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.15.16' (ED25519) to the list of known hosts.
(vmuser2@10.0.15.16) Password for vmuser2@Law Jie Po 01:
FreeBSD 14.3-RELEASE (GENERIC) releng/14.3-n271432-8c9ce319fef7

Welcome to FreeBSD!

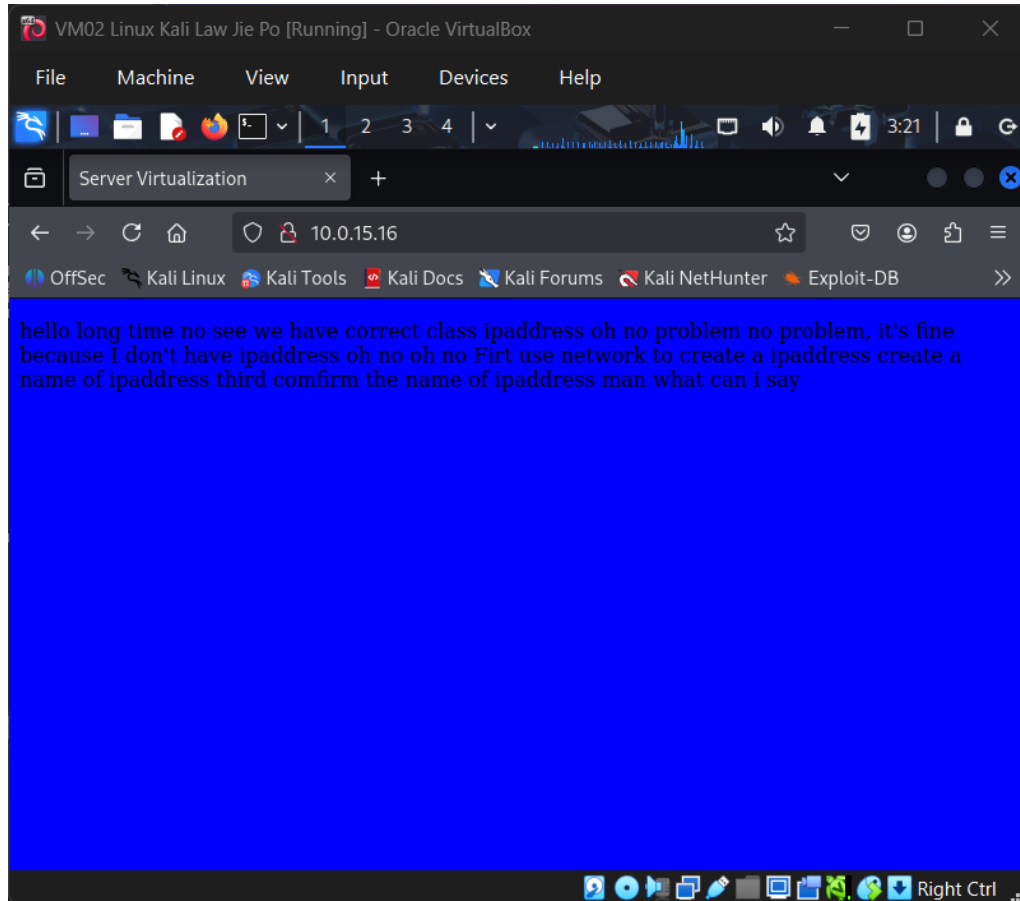
Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories:  https://www.FreeBSD.org/security/
FreeBSD Handbook:     https://www.FreeBSD.org/handbook/
FreeBSD FAQ:          https://www.FreeBSD.org/faq/
Questions List:       https://www.FreeBSD.org/lists/questions/
FreeBSD Forums:       https://forums.FreeBSD.org/

Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with:  pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.
```

5) Browse VM01 website from VM02 – 5m

- i. **Snapshot the browser in VM02 once you surf the website.**

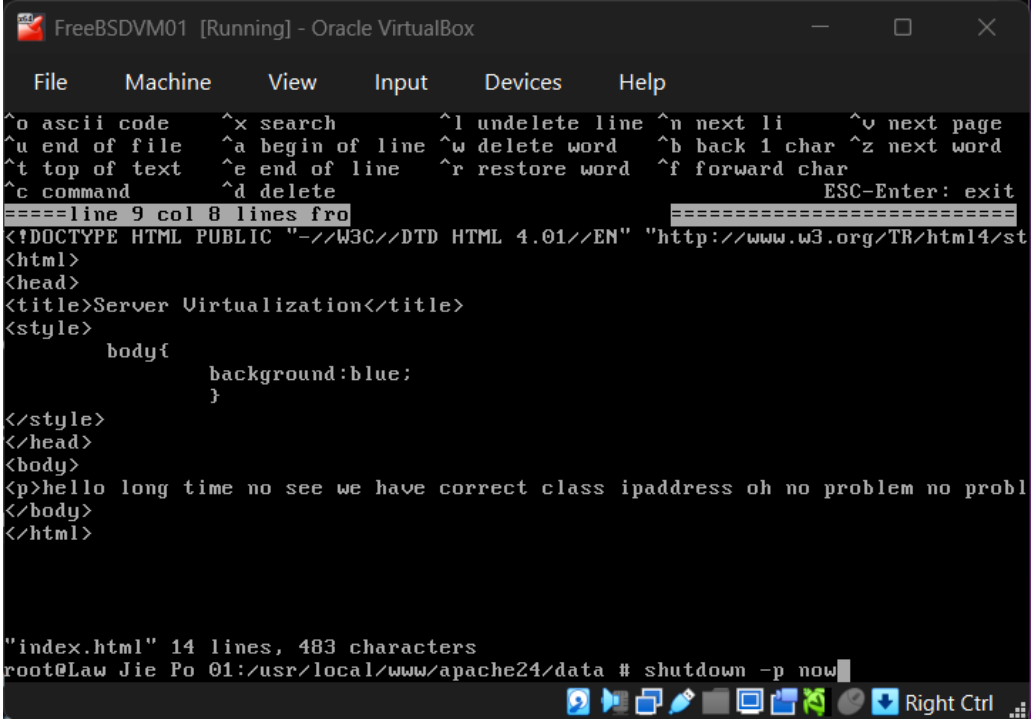
Use `http://ipaddress(10.0.15.16)`



6) Properly power of VM01 & VM02 – 5m

i. Snapshot the commands that you use

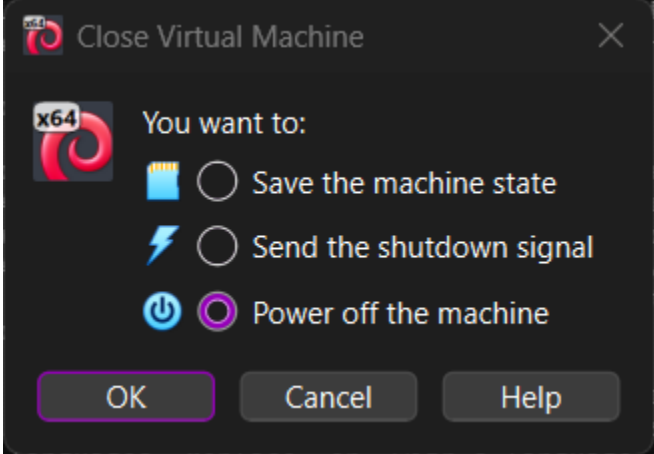
shutdown -p now in vm01

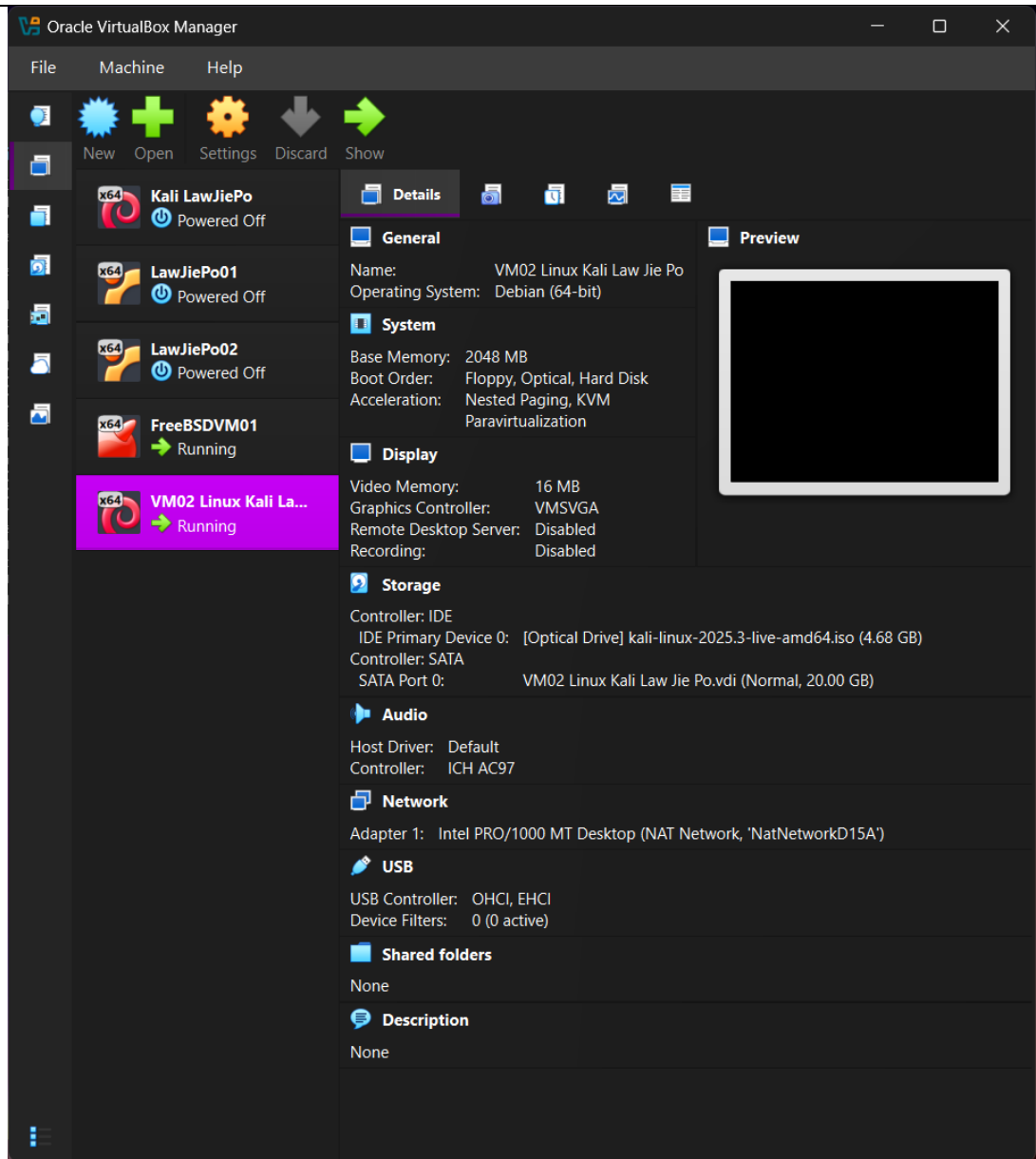


```
FreeBSDVM01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
^o ascii code ^x search ^l undelete line ^n next li ^v next page
^u end of file ^a begin of line ^w delete word ^b back 1 char ^z next word
^t top of text ^e end of line ^r restore word ^f forward char
^c command ^d delete ESC-Enter: exit
====line 9 col 8 lines fro=====
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/st
<html>
<head>
<title>Server Virtualization</title>
<style>
    body{
        background:blue;
    }
</style>
</head>
<body>
<p>hello long time no see we have correct class ipaddress oh no problem no probl
</body>
</html>

"index.html" 14 lines, 483 characters
root@Law Jie Po 01:/usr/local/www/apache24/data # shutdown -p now
```

shutdown in vm02 just esc and then power off the machine

	
7) Snapshot your VirtualBox showing only VM01 & VM02 – 2m	
i.	Snapshot the VirtualBox interface



--	--